

Finite-Sample Reconstruction: Rates, Witnesses, and the Honest Bridge

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1 Setup and Notation

1.1 The measure-preserving system

Let $(X, \mathcal{B}, \mu)$ be a probability space where X is a compact smooth d -dimensional Riemannian manifold and μ is a Borel probability measure with smooth, everywhere-positive density $\rho : X \rightarrow \mathbb{R}_{>0}$ satisfying

$$\rho_{\min} := \operatorname{ess\,inf}_X \rho > 0.$$

Let $T : X \rightarrow X$ be a μ -preserving measurable map. Paper IV makes no injectivity or invertibility assumption on T beyond what is stated in each hypothesis.

1.2 The observable and the delay map

Fix an observable $h \in L^\infty(\mu)$ with $\|h\|_{L^\infty} =: a < \infty$. For each lag depth $L \geq 0$ define the *delay map*

$$\Phi_h^{(L)} : X \rightarrow \mathbb{R}^{L+1}, \quad \Phi_h^{(L)}(x) = (h(x), h(Tx), \dots, h(T^L x)).$$

The *observable σ -algebra at lag L* is

$$\mathcal{O}_h^{(L)} := \sigma(\Phi_h^{(L)}) = \sigma\{h \circ T^k : 0 \leq k \leq L\}.$$

The *delay-polynomial class* of degree $D \geq 1$ is

$$\mathcal{A}_h^{(L)} := \{p \circ \Phi_h^{(L)} : p \text{ is a polynomial of degree } \leq D \text{ on } \mathbb{R}^{L+1}\} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu).$$

Since $\Phi_h^{(L)}(X) \subseteq [-a, a]^{L+1}$, all elements of $\mathcal{A}_h^{(L)}$ are bounded.

1.3 The approximation error

The central quantity of this paper is the σ -algebra approximation error introduced in Section 2:

$$\delta(L) := \delta(\mathcal{O}_h^{(L)}) = \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \Delta E).$$

This measures how much of $\mathcal{B}$ remains outside $\mathcal{O}_h^{(L)}$. Three properties, proved in Papers I–III, are used here without proof:

- (a) *Monotonicity:* $\delta(L)$ is non-increasing in L .
- (b) *Reconstruction:* $\delta(L) \rightarrow 0$ if and only if $\bigcup_L \mathcal{O}_h^{(L)}$ generates $\mathcal{B}$ modulo μ -null sets.
- (c) *Takens regime:* if (T, h) satisfy (G1)–(G2) below and X is a smooth manifold, then $\delta(L) = 0$ for all $L \geq L_0$ where $L_0 \leq d - 1$.

1.4 The data model

We observe a time series $x_1, x_2, \dots, x_n$ which is an orbit segment of T : $x_i = T^{i-1}x_1$ for some initial condition $x_1 \in X$ drawn from μ . The empirical measure is $\mu_n := \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$.

Remark 1.1 (What we do not assume). We do not assume the orbit is dense, that T is ergodic, or that the time average equals the space average. The convergence results in this paper hold under the mixing hypothesis stated explicitly in each result; ergodicity is neither assumed globally nor used silently.

1.5 Structural hypotheses

The following hypotheses are referenced throughout. They are not assumed globally; each theorem states exactly which subset it uses.

- (G1) *No short periodic orbits*: T has no periodic orbit of period $\leq L$ in $\text{supp}(\mu)$. This ensures the orbit points $x, Tx, \dots, T^L x$ are distinct μ -a.e., which is used in the Sard–Smale argument for full rank of $\Phi_h^{(L)}$.
- (G2) *Non-degenerate observable*: $h \in C^2(X)$ with $dh \neq 0$ μ -a.e. This places h in the Sard–Smale regular-value residual set and ensures $\Phi_h^{(L)}$ has rank $\min(L+1, d)$ generically.
- (SS) *Separation-stability*: $\inf_{x \in X} \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, where $\sigma_{\min}$ denotes the smallest singular value. This is a uniform lower Lipschitz bound on $\Phi_h^{(L)}$, strictly stronger than (G1)–(G2) which give full rank only μ -a.e. Condition (SS) is used only in the Dynamics and Conjunction theorems (Sections ?? and ??).

Remark 1.2 (Gap between (G1)–(G2) and (SS)). Hypotheses (G1)–(G2) yield $D\Phi_h^{(L)}$ full rank μ -a.e. (a set of full measure), but allow the rank to drop on a μ -null set. (SS) requires uniform positivity of $\sigma_{\min}$ across all of X , including the μ -null exceptional set. The gap is genuine and cannot be closed without additional hypotheses.

1.6 The empirical witness

The empirical analogue of $\delta(L)$ is

$$\hat{\delta}(L, n) := \sup_{p \in \mathcal{P}_D^{(L)}, \|p\|_{L^2(\mu)} \leq 1} \|p - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu_n)}^2,$$

where $\mathcal{P}_D^{(L)}$ is the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$ (normalised in $L^2(\mu)$), and $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson kernel regression estimator (Definition 3.6).

The main result of Section 3 is that $\hat{\delta}(L, n)$ concentrates around $\delta(L)$ at rate $\varepsilon_n = C n^{-2s/(2s+d)}$ under exponential mixing, provided $d > 2s$.

2 The Bias Bound and the Algebra Side

2.1 The bias bound

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathfrak{m} \subseteq \mathcal{B}$ a sub- σ -algebra. Define the σ -algebra approximation error

$$\delta(\mathfrak{m}) := \sup_{S \in \mathcal{B}} \inf_{E \in \mathfrak{m}} \mu(S \triangle E). \quad (1)$$

This measures how well $\mathfrak{m}$ approximates $\mathcal{B}$: it is zero if and only if $\mathfrak{m} = \mathcal{B}$ modulo μ -null sets.

Lemma 2.1 (Bias bound). *For every $f \in L^\infty(\mu)$,*

$$\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2(\mu)} \leq 2\|f\|_{L^\infty} \cdot \delta(\mathfrak{m})^{1/2}.$$

Proof. Step 1 (Indicators). Fix $S \in \mathcal{B}$. By definition of $\delta(\mathfrak{m})$, there exists $E \in \mathfrak{m}$ with $\mu(S \triangle E) \leq \delta(\mathfrak{m})$. Since $\mathbb{1}_E \in L^2(X, \mathfrak{m}, \mu)$ and $\mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]$ is the L^2 -best $\mathfrak{m}$ -measurable approximation to $\mathbb{1}_S$,

$$\|\mathbb{1}_S - \mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]\|_{L^2}^2 \leq \|\mathbb{1}_S - \mathbb{1}_E\|_{L^2}^2 = \mu(S \triangle E) \leq \delta(\mathfrak{m}).$$

Step 2 (Bounded f via layer cake). For $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \leq M$, the layer cake representation gives $f = \int_{-M}^M \mathbb{1}_{S_t} dt$ where $S_t = \{x : f(x) > t\}$. Since $\mathbb{E}[\cdot \mid \mathfrak{m}]$ is linear and continuous in L^2 ,

$$f - \mathbb{E}[f \mid \mathfrak{m}] = \int_{-M}^M (\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]) dt.$$

Applying Minkowski's inequality for integrals and Step 1:

$$\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2} \leq \int_{-M}^M \|\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\|_{L^2} dt \leq 2M \cdot \delta(\mathfrak{m})^{1/2}. \quad \square$$

Remark 2.2. The constant 2 is not optimal but the exponent $\frac{1}{2}$ is sharp: taking $X = [0, 1]$, $\mathfrak{m} = \{\emptyset, X\}$, $f = \mathbb{1}_{[0, 1/2]}$ gives $\delta(\mathfrak{m}) = \frac{1}{2}$ and $\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2} = \frac{1}{2} = \delta(\mathfrak{m})^{1/2}/\sqrt{2}$.

2.2 The three-term decomposition

We now specialise to the delay setting. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $h \in L^\infty(\mu)$. Write $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ for the observable σ -algebra at lag L , where $\Phi_h^{(L)}(x) = (h(x), h(Tx), \dots, h(T^L x))$, and $\mathcal{A}_h^{(L)}$ for the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$. Let $\hat{f}_n^{(L)}$ denote the empirical risk minimiser over $\mathcal{A}_h^{(L)}$.

Lemma 2.3 (Three-term decomposition). *For any $f \in L^\infty(\mu)$, any $L \geq 0$, and any $D \geq 1$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq \underbrace{2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}}_{\text{bias}} + \underbrace{\inf_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}}_{\text{approx.}} + \underbrace{\|g_D^* - \hat{f}_n^{(L)}\|_{L^2(\mu)}}_{\text{statistical}}, \quad (2)$$

where $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ as in (1) and $g_D^* = \operatorname{argmin}_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}$.

Proof. Triangle inequality applied to the decomposition $f - \hat{f}_n^{(L)} = (f - \mathbb{E}[f \mid \mathcal{O}_h^{(L)}]) + (\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g_D^*) + (g_D^* - \hat{f}_n^{(L)})$, with the first term bounded by Lemma 2.1. $\square$

Remark 2.4 (Scope). Lemma 2.1 holds for any sub- σ -algebra $\mathfrak{m}$; no dynamical structure is needed. The dynamical hypothesis enters only through the approximation and statistical terms, which require the image $\Phi_h^{(L)}(X)$ to be a manageable geometric object.

2.3 The observable reverse lower bound

The three-term bound controls the forward error. For the stopping rule to be honest, we also need a reverse bound: when $\delta(L)$ is large, the estimator $\hat{f}_n^{(L)}$ cannot approximate f well, regardless of n .

Lemma 2.5 (Observable reverse lower bound). *There exists a universal constant $c > 0$ such that for any $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \geq \frac{1}{2}$ and any $L \geq 0$:*

$$\|\mathbb{1}_S - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$$

with probability tending to 1 as $n \rightarrow \infty$, where $S \in \mathcal{B}$ is the set achieving the supremum in (1) at lag L and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ is the empirical measure.

Proof. Step A (Population CE gap). Let $S^* \in \mathcal{B}$ achieve the supremum in $\delta(L) = \sup_S \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E)$. For any $E \in \mathcal{O}_h^{(L)}$:

$$\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2 \leq \|\mathbb{1}_{S^*} - \mathbb{1}_E\|_{L^2(\mu)}^2 = \mu(S^* \triangle E) \geq \delta(L) - \varepsilon$$

for any $\varepsilon > 0$. Taking $\varepsilon \rightarrow 0$: $\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)} \geq \delta(L)^{1/2}$.

Since the conditional expectation is the L^2 -projection, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} \geq \delta(L)^{1/2}$ for all $g \in \mathcal{A}_h^{(L)} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu)$.

Step B (Empirical vs population norm). By Hoeffding's inequality and the bounded difference property of μ_n , $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} - O(n^{-1/2})$ with high probability, uniformly over $g \in \mathcal{A}_h^{(L)}$.

Step C (Empirical norm vs $\hat{f}_n^{(L)}$). Since $\hat{f}_n^{(L)}$ minimises empirical L^2 error over $\mathcal{A}_h^{(L)}$, and $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \delta(L)^{1/2} - O(n^{-1/2})$ for all $g \in \mathcal{A}_h^{(L)}$, we have $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C'n^{-s/(2s+d)}$ with high probability, absorbing the $O(n^{-1/2})$ into the statistical rate. $\square$

Remark 2.6 (Closed loop). Lemmas 2.3 and 2.5 together bound $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)}$ from both sides. The upper bound (Lemma 2.3) says the error is at most $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} + C' \cdot n^{-s/(2s+d)}$. The lower bound (Lemma 2.5) says it is at least $\frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$. The detectability threshold is the n at which the statistical floor $C'n^{-s/(2s+d)}$ drops below $\frac{\delta(L)^{1/2}}{2}$: reconstruction becomes detectable at sample size $n \gtrsim \delta(L)^{-(2s+d)/s}$.

3 Concentration of the Empirical Witness

This section proves that $\hat{\delta}(L, n)$ concentrates around $\delta(L)$. The argument has four steps: uniform boundedness of the function class (Lemma 3.1), McDiarmid concentration around the mean (Proposition 3.3), a Glivenko–Cantelli bound for the LLN gap (Lemma 3.5), and the estimator bias via the Nadaraya–Watson kernel regression (Proposition 3.7). The full convergence is then Theorem 3.9.

Throughout this section the hypotheses of Proposition 4.3 hold: (T, h) satisfies (G1)–(G2), X is a compact smooth d -manifold, μ has smooth positive density, and $d > 2s$.

3.1 Uniform boundedness

Let $\mathcal{P}_D^{(L)}$ denote the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x) = (h(x), \dots, h(T^L x))$, normalised so $\|p\|_{L^2(\mu)} \leq 1$. Since $h \in L^\infty$ with $\|h\|_{L^\infty} =: a < \infty$, the delay coordinates lie in the compact box $[-a, a]^{L+1}$.

Lemma 3.1 (Uniform boundedness). *Define*

$$M_{D,L} := \binom{D+L+1}{L+1}^{1/2} \cdot (2D+1)^{1/2} \cdot (2a)^{-(L+1)/2} \cdot \rho_{\min}^{-1/2},$$

where $\rho_{\min} = \text{essinf}_X \rho > 0$ is the infimum of the density of μ . Then for every $p \in \mathcal{P}_D^{(L)}$:

$$\|p\|_{L^\infty} \leq M_{D,L} \cdot \|p\|_{L^2(\mu)}.$$

Proof. Rescale to the unit box by setting $y_i = x_i/a$; polynomials of degree $\leq D$ in $x \in [-a, a]^{L+1}$ correspond to polynomials $\tilde{p}$ of degree $\leq D$ in $y \in [-1, 1]^{L+1}$. Expand $\tilde{p}$ in the tensor-product Legendre basis $\{P_\alpha\}_{|\alpha| \leq D}$, where $P_\alpha(y) = \prod_{i=1}^{L+1} P_{\alpha_i}(y_i)$ and $\|P_k\|_{L^\infty[-1,1]} = 1$. Since $\|P_\alpha\|_{L^\infty} = 1$, Cauchy–Schwarz gives

$$|\tilde{p}(y)| \leq \sum_{|\alpha| \leq D} |c_\alpha| \leq N(D, L+1)^{1/2} \left(\sum_{|\alpha| \leq D} |c_\alpha|^2 \right)^{1/2},$$

where $N(D, L+1) = \binom{D+L+1}{L+1}$ counts the basis elements. From the L^2 orthogonality $\|P_\alpha\|_{L^2[-1,1]}^2 = \prod_i (2\alpha_i + 1)^{-1}$, the coefficient norm satisfies

$$\sum_{|\alpha| \leq D} |c_\alpha|^2 \leq (2D+1) \|\tilde{p}\|_{L^2}^2,$$

since the maximum of $\prod_i (2\alpha_i + 1)$ over $|\alpha| \leq D$ is $2D+1$, achieved at $\alpha = (D, 0, \dots, 0)$ — the constraint $|\alpha| \leq D$ forces all weight onto one coordinate, giving an L -independent maximum. Converting back via $\|\tilde{p}\|_{L^2[-1,1]^{L+1}} = (2a)^{-(L+1)/2} \|p\|_{L^2(\text{vol})}$ and $\|p\|_{L^2(\text{vol})} \leq \rho_{\min}^{-1/2} \|p\|_{L^2(\mu)}$ yields the result. $\square$

Remark 3.2. The L -dependence of $M_{D,L}$ is entirely in $N(D, L+1)^{1/2} \sim L^{D/2}/\sqrt{D!}$ for $L \gg D$. The factor $(2D+1)^{1/2}$ depends only on degree. For fixed $D = \lceil s \rceil$ and $\|h\|_{L^\infty} < 1$ (so $2a < 1$), $M_{D,L}$ decays exponentially in L ; for $\|h\|_{L^\infty} \geq \frac{1}{2}$ (equivalently $2a \geq 1$), $M_{D,L}$ grows polynomially in L . The subsequent arguments require $\|h\|_{L^\infty} \geq \frac{1}{2}$.

3.2 Concentration around the mean

Define the function class

$$\mathcal{F}_{L,D} = \left\{ x \mapsto (p(x) - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}](x))^2 : p \in \mathcal{P}_D^{(L)}, \|p\|_{L^2(\mu)} \leq 1 \right\},$$

where $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson estimator (Definition 3.6). Then $\hat{\delta}(L, n) = \sup_{f \in \mathcal{F}_{L,D}} \frac{1}{n} \sum_{i=1}^n f(x_i)$.

Proposition 3.3 (Concentration, $\star$). *Assume $d > 2s$. There exist constants $c, C > 0$ depending only on $(d, s, \|h\|_{L^\infty})$ such that for all $t > 0$ and all $n \geq 1$:*

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > t) \leq 2 \exp\left(-\frac{c n t^2}{M_{D,L}^4 \cdot D^d}\right). \quad (\star)$$

In particular, setting $t = \varepsilon_n/2$ with $\varepsilon_n = C' n^{-2s/(2s+d)}$:

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > \varepsilon_n/2) \leq 2 \exp\left(-c' n^{(d-2s)/(2s+d)}\right) \rightarrow 0.$$

Proof. Step 1 (Boundedness). By Lemma 3.1, every $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$ satisfies $\|p\|_{L^\infty} \leq M_{D,L}$. Hence every $f = (p - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}])^2 \in \mathcal{F}_{L,D}$ satisfies $\|f\|_{L^\infty} \leq 4M_{D,L}^2 =: B_{L,D}^2$.

Step 2 (McDiarmid). Replacing one observation x_i with x'_i changes $\hat{\delta}(L, n)$ by at most $B_{L,D}^2/n$. By the McDiarmid bounded-difference inequality:

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > t) \leq 2 \exp\left(-\frac{2n^2 t^2}{n \cdot (B_{L,D}^2/n)^2}\right) = 2 \exp\left(-\frac{n t^2}{2B_{L,D}^4/n}\right).$$

Step 3 (Rademacher bias). By the symmetrisation inequality [?]:

$$\mathbb{E}[\hat{\delta}(L, n)] - \delta(L) \leq 2 \mathfrak{R}_n(\mathcal{F}_{L,D}).$$

By the Ledoux–Talagrand contraction lemma, since $f = (p - \cdot)^2$ is $2B_{L,D}$ -Lipschitz in the second argument:

$$\mathfrak{R}_n(\mathcal{F}_{L,D}) \leq 2B_{L,D} \mathfrak{R}_n(\mathcal{P}_D^{(L)}).$$

In the Takens regime the image $\Phi_h^{(L)}(X)$ is a d -dimensional submanifold, so the VC dimension of $\mathcal{P}_D^{(L)}$ restricted to $\Phi_h^{(L)}(X)$ is $O(D^d)$ [?]. Hence $\mathfrak{R}_n(\mathcal{P}_D^{(L)}) \leq C\sqrt{D^d/n}$, giving $\mathfrak{R}_n(\mathcal{F}_{L,D}) \leq C M_{D,L} \sqrt{D^d/n}$.

Step 4 (Combined). Combining Steps 2–3 and absorbing constants:

$$P(|\hat{\delta}(L, n) - \delta(L)| > t) \leq 2 \exp\left(-\frac{c n t^2}{M_{D,L}^4 D^d}\right).$$

Setting $t = \varepsilon_n/2 = C' n^{-2s/(2s+d)}/2$, the exponent becomes

$$c n \varepsilon_n^2 / (M_{D,L}^4 D^d) \propto n^{1-4s/(2s+d)} = n^{(d-2s)/(2s+d)}.$$

This tends to $+\infty$ if and only if $d > 2s$, proving the final claim. $\square$

Remark 3.4 (The $d > 2s$ hypothesis). The condition $d > 2s$ is necessary for the concentration to be useful at the rate ε_n . It is not an artefact: when $d \leq 2s$, the function class $\mathcal{F}_{L,D}$ is too rich for n samples to resolve the fine structure of $\delta(L)$ at the required scale. The fix via localised Rademacher complexity [?] is technically available but left to future work; Paper IV takes $d > 2s$ as a standing hypothesis.

3.3 Source 1: the LLN gap

Lemma 3.5 (LLN gap). *With notation as above:*

$$\sup_{p \in \mathcal{P}_D^{(L)}} \left| \|p - \mathbb{E}[p | \Phi_h^{(L)}]\|_{L^2(\mu_n)}^2 - \|p - \mathbb{E}[p | \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \right| \leq C \sqrt{D^d/n}$$

almost surely as $n \rightarrow \infty$, where C depends on $(d, D, M_{D,L})$.

Proof. The class $\{(p - \mathbb{E}[p | \Phi_h^{(L)}])^2 : p \in \mathcal{P}_D^{(L)}\}$ is a VC class of dimension $O(D^d)$. The uniform Glivenko–Cantelli theorem [?] gives a.s. convergence of the empirical mean to the population mean, at rate $O(\sqrt{D^d/n})$. $\square$

3.4 Source 2: estimator bias

The estimator $\hat{\mathbb{E}}[p | \Phi_h^{(L)}]$ is the Nadaraya–Watson (NW) kernel regression estimator.

Definition 3.6 (Nadaraya–Watson estimator). For bandwidth $h_n > 0$ and kernel $K : \mathbb{R}^{L+1} \rightarrow \mathbb{R}_{\geq 0}$ (compactly supported, symmetric, $\int K = 1$):

$$\hat{\mathbb{E}}[p | \Phi_h^{(L)}(x)] := \frac{\sum_{j=1}^n K\left(\frac{\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x_j)}{h_n}\right) p(x_j)}{\sum_{j=1}^n K\left(\frac{\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x_j)}{h_n}\right)}.$$

Proposition 3.7 (Source 2 estimator bias). *Assume:*

- (i) $T, h \in C^r$ with $r \geq 2$, μ smooth positive density ρ with $\rho_{\min} > 0$.

(ii) $\Phi_h^{(L)}$ has full rank μ -a.e. (follows from (G1)–(G2) and Lemma 4.2).

(iii) T is exponentially mixing with rate $\lambda > 0$ and $\|h\|_{L^\infty} \geq \frac{1}{2}$.

(iv) The NW bandwidth satisfies $h_n \sim n^{-1/(2\beta+d_{\text{eff}}(L))}$ with $d_{\text{eff}}(L) = \min(L+1, d)$ and $\beta \geq 1$.

Then there exists $C_2(D, L) > 0$ such that for all $p \in \mathcal{P}_D^{(L)}$:

$$\mathbb{E}\|\hat{\mathbb{E}}[p | \Phi_h^{(L)}] - \mathbb{E}[p | \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \leq C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\text{eff}}(L))},$$

uniformly in $p \in \mathcal{P}_D^{(L)}$, where $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$ and A is a constant depending on $(\beta, d_{\text{eff}}(L), \rho_{\min})$.

Proof. The proof is a four-link chain.

Link 1 (Smooth disintegration). Under conditions (i)–(ii), the Rokhlin disintegration $\mu = \int \mu_z d((\Phi_h^{(L)})_* \mu)(z)$ has fibre measures μ_z supported on the smooth fibres $F_z = \{\Phi_h^{(L)} = z\}$. An implicit-function-theorem argument (three sub-steps: transfer to common surface via a C^{r-1} diffeomorphism; bound weight variation; bound Jacobian factor) gives:

$$\|\mu_z - \mu_{z'}\|_{\text{TV}} \leq C_1 |z - z'|^\beta, \quad \beta = 1 \text{ (Lipschitz floor)},$$

where C_1 depends on $(\|T\|_{C^r}, \|h\|_{C^r}, \rho_{\min})$. (The optimal exponent $\beta = r - 1$ follows from the full Taylor expansion of the fibre weight; the floor $\beta = 1$ is sufficient for convergence.)

Link 2 (Bernstein–Markov). From Link 1 and Lemma 3.1: for $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$,

$$|\mathbb{E}[p | \Phi_h^{(L)} = z] - \mathbb{E}[p | \Phi_h^{(L)} = z']| = \left| \int p d\mu_z - \int p d\mu_{z'} \right| \leq M_{D,L} \cdot C_1 \cdot |z - z'|^\beta.$$

Hence $z \mapsto \mathbb{E}[p | \Phi_h^{(L)} = z]$ is Hölder(β) with constant $\Lambda_{D,L} := M_{D,L} C_1$, uniformly over $p \in \mathcal{P}_D^{(L)}$.

Link 3 (NW bias-variance balance). Standard NW theory [?] for a Hölder(β) regression function on a $d_{\text{eff}}(L)$ -dimensional domain gives:

$$\mathbb{E}\|\hat{\mathbb{E}}[p | \Phi_h^{(L)}] - \mathbb{E}[p | \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \leq \Lambda_{D,L}^2 h_n^{2\beta} + \frac{C_K M_{D,L}^2}{n h_n^{d_{\text{eff}}(L)} f_{\min}},$$

where C_K depends on the kernel K and $f_{\min}$ is the minimum of the marginal density of $(\Phi_h^{(L)})_* \mu$. The $M_{D,L}^2$ factor appears in both terms. Balancing: $\Lambda_{D,L}^2 h_n^{2\beta} = C_K M_{D,L}^2 / (n h_n^{d_{\text{eff}}(L)} f_{\min})$ gives $h_n^* \sim A \cdot n^{-1/(2\beta+d_{\text{eff}}(L))}$ where A depends only on $(\beta, d_{\text{eff}}(L), C_K, f_{\min}, C_1)$ — not on D, L , or p . The $M_{D,L}^2$ factors cancel from both sides of the balance equation.

Link 4 (Uniformity). Substituting h_n^* , both terms contribute $C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\text{eff}}(L))}$ with $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$. Since the bound is p -free, it holds uniformly over $\mathcal{P}_D^{(L)}$. $\square$

Remark 3.8 (Scope). Proposition 3.7 is proved under exponential mixing with $\|h\|_{L^\infty} \geq \frac{1}{2}$. The polynomial mixing case and the regime $\|h\|_{L^\infty} < \frac{1}{2}$ require a different estimator and are left open.

3.5 Full convergence

Theorem 3.9 (Full convergence $\hat{\delta}(L, n) \rightarrow \delta(L)$). *Under the hypotheses of Proposition 3.7 and with $d > 2s$:*

$$\mathbb{E}|\hat{\delta}(L, n) - \delta(L)| \leq C \sqrt{D^d/n} + C_2(D, L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\text{eff}}(L))}$$

and $P(|\hat{\delta}(L, n) - \delta(L)| > \varepsilon_n) \rightarrow 0$ exponentially, where $\varepsilon_n = C' n^{-2s/(2s+d)}$.

Proof. Decompose:

$$|\hat{\delta}(L, n) - \delta(L)| \leq \underbrace{|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]|}_{\text{concentration, } (\star)} + \underbrace{|\mathbb{E}[\hat{\delta}(L, n)] - \delta(L)|}_{\text{bias}}$$

The bias splits as Source 1 + Source 2: Source 1 is bounded by $C\sqrt{D^d/n}$ (Lemma 3.5); Source 2 is bounded by $C_2(D, L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\text{eff}}(L))}$ (Proposition 3.7). The concentration term is bounded by $(\star)$ (Proposition 3.3). Combining at $t = \varepsilon_n/2$ gives the probability statement. $\square$

4 The Algebra Theorem

This section assembles the results of Sections 2 and 3 into the Algebra Theorem: the data-driven stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ without knowing the oracle lag $L^*(n)$ or the mixing rate of T .

4.1 Effective dimension and the piecewise rate

Definition 4.1 (Effective dimension). The *effective dimension at lag L* is

$$d_{\text{eff}}(L) := \text{rank}(D\Phi_h^{(L)}|_x) \quad \text{at a generic point } x \in X.$$

(The rank is constant on an open dense set by the rank theorem, so this is well-defined μ -a.e.)

Lemma 4.2 (d_{eff} step function). *Under (G1)–(G2) with X a compact smooth d -manifold and μ smooth positive density:*

$$d_{\text{eff}}(L) = \min(L + 1, d) \quad \text{for all } L \geq 0.$$

Proof. The upper bound $d_{\text{eff}}(L) \leq \min(L + 1, d)$ holds because $D\Phi_h^{(L)}|_x$ is a linear map from $T_x X$ (dimension d) to $\mathbb{R}^{L+1}$.

For the lower bound we use the Sard–Smale theorem (Smale [?] framework) applied to two failure maps.

Case $L + 1 \leq d$. Rank $< L + 1$ at x means the cotangent vectors $\omega_j(x) := (DT^j|_x)^*(dh|_{T^j x})$ are linearly dependent. Define the failure map

$$\Psi : C^2(X) \times X \times \mathbb{P}^L \rightarrow T^*X, \quad (h, x, [\lambda]) \mapsto d\left(\sum_j \lambda_j h \circ T^j\right)|_x.$$

By (G1), the orbit points $T^j x$ are distinct μ -a.e., so bump functions at $T^j x$ can prescribe any cotangent vector at x independently. Hence $D_h \Psi$ is surjective onto $T_x^* X$ (dimension d), and $\Psi^{-1}(0)$ is a Banach submanifold of codimension d . The projection to $C^2(X)$ is Fredholm of index $d + L - d = L$, so by Sard–Smale the regular values form a residual set $R \subset C^2(X)$. For $h \in R$, the failure locus in X has dimension $\leq L < d$, hence μ -measure zero.

By (G2), $dh \neq 0$ μ -a.e., which forces every critical point of $F_\lambda = \sum_j \lambda_j h \circ T^j$ to lie in the codimension- d submanifold $S_\lambda = \{dh_x \in \text{span}\{\omega_1(x), \dots, \omega_L(x)\}\}$ (a finite set for each $[\lambda]$, since X is d -dimensional). Hence h is a regular value of π , i.e., $h \in R$.

Case $L + 1 > d$. Immersion failure requires $\exists v \neq 0$ with $D\Phi_h^{(L)}|_x \cdot v = 0$. Define

$$\Xi : C^2(X) \times (TX \setminus 0) \rightarrow \mathbb{R}^{L+1}, \quad (h, (x, v)) \mapsto (dh_{T^j x}(DT^j|_x \cdot v))_{j=0}^L.$$

By (G1), $D_h \Xi$ is surjective onto $\mathbb{R}^{L+1}$, so $\Xi^{-1}(0)$ has codimension $L + 1$ and the projection to $C^2(X)$ is Fredholm of index $2d - 1 - (L + 1) = 2d - L - 2$. For $L \geq d$: $2d - L - 2 \leq d - 2 < d - 1$, so the failure locus in X has dimension $< d - 1$ and μ -measure zero. Condition (G2) places h in the regular-value set by the same argument as Case 1. $\square$

Proposition 4.3 (Piecewise pre-Takens rate). *Assume (G1)–(G2), X compact smooth d -manifold, μ smooth positive density, $f \in \text{H\"older}(s) \cap L^\infty(\mu)$, $d > 2s$, and $T \in C^r$ with $r \geq 2$. Let $\hat{f}_n^{(L)}$ be the empirical risk minimiser over $\mathcal{A}_h^{(L)}$ with $D = D^*(n, L)$ chosen to balance approximation and statistical terms. Then for each $L \geq 0$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq 2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d_{\text{eff}}(L))}$$

with probability tending to 1, where $d_{\text{eff}}(L) = \min(L+1, d)$. In the Takens regime ($L \geq d-1$): $d_{\text{eff}}(L) = d$ and the rate locks at $n^{-s/(2s+d)}$.

Proof. Apply the three-term decomposition (Lemma 2.3). The bias term is bounded by $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}$ (Lemma 2.1). By Lemma 4.2, $\Phi_h^{(L)}(X)$ is a $d_{\text{eff}}(L)$ -dimensional smooth submanifold. On this submanifold, $\mathbb{E}[f \mid \mathcal{O}_h^{(L)}](x) = g^*(\Phi_h^{(L)}(x))$ for some $g^* \in \text{H\"older}(s)$; standard polynomial approximation on a $d_{\text{eff}}(L)$ -dimensional smooth domain gives approximation error $\leq C_{\text{approx}} \cdot D^{-s/d_{\text{eff}}(L)}$. The statistical term is $\leq C_{\text{stat}}(D^{d_{\text{eff}}(L)}/n)^{1/2}$ by Proposition 3.3 with d replaced by $d_{\text{eff}}(L)$. Balancing: $D^* \sim n^{d_{\text{eff}}(L)/(2s+d_{\text{eff}}(L))}$ and both terms are $O(n^{-s/(2s+d_{\text{eff}}(L))})$. $\square$

4.2 The oracle lag and the elbow stopping rule

Definition 4.4 (Oracle lag). The *oracle lag* is

$$L^*(n) := \min\{L \geq 0 : \delta(L) \leq C_0 \cdot n^{-2s/(2s+d)}\},$$

where $C_0 > 0$ is a constant depending on $\|f\|_{L^\infty}$. At $L = L^*(n)$, the bias term $2\|f\|_{L^\infty} \cdot \delta(L^*)^{1/2}$ is $O(n^{-s/(2s+d)})$, matching the statistical floor.

Remark 4.5 (Post-saturation self-consistency). Under exponential mixing $\delta(L) \leq Ae^{-\lambda L}$, balancing $e^{-\lambda L/2} \sim n^{-s/(2s+d)}$ gives $L^*(n) \sim \frac{2s}{\lambda(2s+d)} \log n$. For large n , $L^*(n) \gg d-1$, so the oracle lag lies in the Takens regime and the rate $n^{-s/(2s+d)}$ is self-consistent. Under polynomial mixing $\delta(L) \leq AL^{-\alpha}$, $L^*(n) \sim n^{2s/(\alpha(2s+d))} \rightarrow \infty$, also post-saturation.

The oracle lag $L^*(n)$ requires knowledge of the mixing rate, which is unobservable. The empirical witness $\hat{\delta}(L, n)$ sidesteps this: it measures reconstruction quality directly from data.

Definition 4.6 (Elbow stopping rule). Fix a tolerance sequence $\tau_n > 0$. The *elbow stopping rule* is

$$\hat{L}^* := \min\{L \geq 0 : \hat{\delta}(L+1, n) - \hat{\delta}(L, n) > -\tau_n\}.$$

Theorem 4.7 (Elbow location). *Assume the hypotheses of Proposition 4.3 and exponential mixing $\delta(L) \leq Ae^{-\lambda L}$ with resolvability constant $C_\lambda := (1 - e^{-\lambda})Ae^\lambda/C' > 4$. Set $\tau_n = \frac{C_\lambda}{2} \cdot C' \cdot n^{-2s/(2s+d)}$. Then*

$$P(\hat{L}^* = L^*(n)) \rightarrow 1 \quad \text{exponentially in } n.$$

Proof. We show the stopping criterion fires at $L = L^*(n)$ (no overshoot) and not before (no early stop).

No overshoot: $\hat{L}^* \leq L^*(n)$. At $L = L^*(n)$, the population increment satisfies $|\delta(L^*(n) + 1) - \delta(L^*(n))| \leq \delta(L^*(n)) \leq C_0 n^{-2s/(2s+d)}$ (non-increasing δ). By Theorem 3.9, $|\hat{\delta}(L, n) - \delta(L)| \leq \varepsilon_n = C' n^{-2s/(2s+d)}$ whp. The empirical increment satisfies $\hat{\delta}(L^* + 1, n) - \hat{\delta}(L^*, n) \geq -C_0 n^{-2s/(2s+d)} - 2\varepsilon_n > -\tau_n$ whp, so the criterion fires.

No early stop: $\hat{L}^* \geq L^*(n)$. For $L < L^*(n)$, exponential mixing gives population decrement $\delta(L) - \delta(L+1) \geq \Delta(L) \geq C_\lambda \cdot C' n^{-2s/(2s+d)}$. Tracking via Theorem 3.9:

$$\hat{\delta}(L, n) - \hat{\delta}(L+1, n) \geq C_\lambda C' n^{-2s/(2s+d)} - 2\varepsilon_n = (C_\lambda - 2)C' n^{-2s/(2s+d)}.$$

Since $C_\lambda > 4$: $(C_\lambda - 2)C' > \tau_n \cdot 2/C_\lambda \cdot C_\lambda = 2\tau_n/C_\lambda \cdot C_\lambda$, and specifically $(C_\lambda - 2)C' n^{-2s/(2s+d)} \geq \tau_n$ when $C_\lambda - 2 \geq C_\lambda/2$, i.e., $C_\lambda \geq 4$. So the stopping criterion does *not* fire for $L < L^*(n)$, and $\hat{L}^* \geq L^*(n)$.

Remark 4.8 (Scope of the elbow theorem). The resolvability condition $C_\lambda > 4$ is not an artefact: when $C_\lambda \leq 4$ the elbow drop at $L^*(n)$ merges with the noise floor ε_n and the two are indistinguishable from data. For polynomial mixing, $\Delta(L^*(n)) \sim \varepsilon_n/L^*(n) \rightarrow 0$ relative to ε_n , so the elbow drops vanish asymptotically and the theorem fails. A separate argument (e.g. integrated elbow) is needed for that case and is not proved here. □

4.3 The Algebra Theorem

Theorem 4.9 (Algebra Theorem). *Assume:*

- (i) *Hypotheses of Proposition 4.3: (G1)–(G2), X compact smooth d -manifold, μ smooth positive density, $f \in \text{Hölder}(s) \cap L^\infty(\mu)$, $d > 2s$, $T \in C^r$, $r \geq 2$.*
- (ii) *Exponential mixing: $\delta(L) \leq Ae^{-\lambda L}$ for some $A, \lambda > 0$.*
- (iii) *Resolvability: $C_\lambda > 4$.*
- (iv) *Exponential mixing for the estimator bias (Proposition 3.7) and $\|h\|_{L^\infty} \geq \frac{1}{2}$.*

Let $\hat{L}^*$ be the elbow stopping rule (Definition 4.6) with $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$. Then

$$\|f - \hat{f}_n^{(\hat{L}^*)}\|_{L^2(\mu)} = O_p(n^{-s/(2s+d)}).$$

The rate $n^{-s/(2s+d)}$ is minimax-optimal for Hölder(s) functions on a d -dimensional domain. No oracle input (mixing rate, $L^*(n)$, Sobolev index s) is required.

Proof. On the event $\{\hat{L}^* = L^*(n)\}$, which has probability tending to 1 by Theorem 4.7:

$$\begin{aligned} \|f - \hat{f}_n^{(\hat{L}^*)}\|_{L^2(\mu)} &= \|f - \hat{f}_n^{(L^*(n))}\|_{L^2(\mu)} \\ &\leq 2\|f\|_{L^\infty} \cdot \delta(L^*(n))^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d)} \\ &\leq 2\|f\|_{L^\infty} \cdot (C_0 n^{-2s/(2s+d)})^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d)} \\ &= O(n^{-s/(2s+d)}). \end{aligned} \quad \square$$

Remark 4.10 (What is and is not required). The proof uses: (a) Lemma 4.2 to lock the rate at $n^{-s/(2s+d)}$ once $L \geq d - 1$; (b) Theorem 3.9 to track $\delta(L)$ from data; (c) Theorem 4.7 to locate $L^*(n)$ without knowing the mixing rate; (d) Proposition 4.3 to bound the error at $L^*(n)$. The conditions $C_\lambda > 4$ and $\|h\|_{L^\infty} \geq \frac{1}{2}$ are load-bearing: removing them makes either the elbow unresolvable or the bias estimate (Proposition 3.7) unavailable. Polynomial mixing is not covered.